

# Dean Kuhn

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## Summary

Computer science graduate with a focus on data engineering and analytics. Built four end-to-end portfolio projects spanning automated ETL/ELT pipelines, ML-driven forecasting, and algorithm optimization, including two pipelines on scheduled GitHub Actions, a full Postgres/Snowflake/dbt cloud data warehouse stack, and a configurable Genetic Algorithm optimizer. Deployed on AWS EC2 with systemd services and nightly cron orchestration. Eager to apply hands-on engineering skills in a junior data engineering, analytics engineering, or data analysis role.

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## Projects

### [KitchenSync Food Forecasting System](#) · 2026

*Tech: Python · FastAPI · PostgreSQL (Neon) · Snowflake · dbt Core · LightGBM · scikit-learn · Streamlit · Docker · AWS EC2*

- Architected an end-to-end ELT pipeline simulating a retail kitchen production system: async FastAPI ingest layer receiving live POS events from 12 stores, per-store schema isolation across Neon Postgres, nightly extract to Snowflake data warehouse, three-layer dbt Core transformation (staging → intermediate → marts), LightGBM forecasting model, and a Streamlit dashboard with 5-minute auto-refresh showing split Kitchen and Chicken production queues.
- Built six dbt mart models in Snowflake covering production targets, item velocity with urgency flags, waste percentage by category, cold-start profiles, stockout summaries, and store-level sales aggregates; mart-on-mart refs enforce dependency ordering throughout the transformation layer.
- Engineered an async POS simulator (httpx + asyncio) using Poisson customer arrivals and a time-of-day RUSH\_CURVE; each store runs as an independent async task with a StoreState class tracking FIFO batch inventory, expiration, and stockout events; simulator resumes from Snowflake watermark on restart.
- Deployed FastAPI ingest API and POS simulator as systemd services on AWS EC2 with auto-restart; nightly cron chains extract → dbt → predict → A/B comparison → git push; GitHub Actions triggers portfolio site rebuild each morning.
- Instrumented a nightly A/B comparison (ML vs. hourly-average baseline) under identical seeded Poisson demand: ML system generates 338,688 predictions at 15-minute slot grain (12 stores × 42 items × 672 weekly slots), cuts stockouts ~40%, and lifts service level +1.6pp at the cost of +3.3pp additional waste.
- Diagnosed two silent data quality bugs: a day\_of\_week convention mismatch between Snowflake EXTRACT and ISO weekday caused 2.37M training rows to carry slot\_quantity=0, producing a near-constant model; a conditional mean bias inflated demand predictions 3–4x at low-traffic stores by averaging only slots with non-zero sales rather than all observed slots.

### [Music Growth Pipeline](#) · 2026

*Tech: Python · PostgreSQL (Neon) · dbt Core (dbt-postgres) · Last.fm API · GitHub Actions · SQL · Power BI*

- Designed and deployed a weekly ingestion pipeline that snapshots listener and playcount data for 7,751 artists from the Last.fm API into a cloud Postgres data warehouse, building longitudinal time-series data for engagement analysis that the API does not natively provide.
- Built a dbt transformation layer (6 staging + 7 mart models) powering cross-sectional and longitudinal analysis; marts build on marts via ref(), including listener\_growth (LAG window function), artist\_growth\_summary, weekly\_growth\_by\_tier, and genre\_growth.
- Extended the schema to capture genre associations (15 genres × 500 artists) and artist similarity networks (~2,000 indie artists, 20 similar artists each), enabling genre-level breakdowns and cross-tier network analysis.
- Core longitudinal finding across 7 weeks of data: underground artists (pages 1000+) show P90 listener growth of 9.16% vs. mainstream 2.75% — a 3x gap concentrated in the tail; automated weekly snapshots push fresh stats to portfolio site via pipeline\_stats.json.
- Built a two-page Power BI report connected live to Neon Postgres, visualizing tier growth trends and genre breakdowns with drillthrough to individual artist detail.

### [Market Sentiment Medallion Pipeline](#) · 2026

*Tech: Python · Playwright · VADER (NLTK) · Pydantic · pandas · PyArrow · Streamlit · GitHub Actions*

- Bronze→Silver→Gold medallion ELT pipeline correlating Reddit sentiment across four subreddits, weighted by a custom controversy heuristic and per-subreddit trust multipliers, with Yahoo Finance price data to detect sentiment-price divergence signals.

## **VRP-TW Genetic Algorithm Routing Optimizer · 2026**

*Tech: Python · Genetic Algorithm · CLI*

- Built a Genetic Algorithm that simultaneously optimizes package-to-truck assignment and delivery sequence using a sentinel chromosome encoding, solving a combined loading and routing problem in a single fitness function rather than two sequential passes.
- Implemented adaptive mutation (rate doubles after 50 stagnant generations), scramble and inversion operators, sentinel-shift mutation to prevent capacity distribution from fossilizing, and gradient deadline penalty ( $\text{minutes\_late} \times 10$ ) to create a smooth fitness landscape over binary pass/fail.
- Configurable CLI supporting custom package counts, truck capacity, refrigeration constraints, and GA hyperparameters; post-run package status lookup by ID or address reconstructed from simulation output without re-running.

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## **Technical Skills**

**Languages:** Python, SQL

**Data & Storage:** PostgreSQL, Snowflake, dbt Core (dbt-postgres, dbt-snowflake), pandas, PyArrow, Parquet

**Cloud & Infra:** AWS EC2, GitHub Actions, Docker, Neon (cloud Postgres), systemd, Git, Linux/WSL

**Ingestion & APIs:** FastAPI, REST APIs (Last.fm), Playwright (headless scraping), asyncio/httpx

**ML & Modeling:** LightGBM, scikit-learn, VADER sentiment analysis

**Visualization:** Streamlit, Plotly, Power BI, Excel

**Concepts:** ETL/ELT pipeline design, medallion architecture, data warehouse modeling, dbt staging/mart patterns, ML inference serving, cloud data platforms

**Exposure:** PySpark, Apache Airflow, Databricks, Azure

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## **Education**

**B.S. Computer Science · Western Governors University · May 2026**

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## **Work Experience**

### **Guest Services & Inventory Coordinator · Kwik Trip · 2021 – Present**

- Tracked inventory and managed ingredient purchasing for kitchen operations in a high-volume food production environment (the same system KitchenSync models), balancing stock levels daily to avoid both shortages and excess waste.
- Ran end-of-shift cash reconciliation and register audits across high daily transaction volume; caught and resolved discrepancies before close.
- First-line troubleshooting for POS terminal failures, network issues, and delivery problems during live shifts with no dedicated IT support on site.
- Trained incoming coworkers on inventory systems, cash handling, and kitchen production workflows; kept operations running smoothly through constant staff turnover.